

Interview coding task

Repository template

For this task you are asked to use the repository template that is attached to this email. The README.md file contains extensive information about the tools used in the template, but here is a short list of instructions to get things running:

1. unzip the archive
2. If you don't have uv, [install uv](#)
3. cd lightning-hydra-uv-template/
4. uv sync – create virtual environment
5. uv run pre-commit install – install pre-commit hooks
6. uv run pre-commit run --all – make sure all checks pass
7. uv run pytest – make sure all tests pass
8. uv run src/train.py – This will start model training. It will run on cpu by default, takes about 1-2 minutes.
9. After the training is done, get the path to the saved model checkpoint inside ./logs/train/runs
10. uv run src/eval.py ckpt_path="path/to/model/checkpoint/file.ckpt"

Please let us know if you encounter any issues using this template.

Task

The objective of this project is to fine-tune a small language model on a small text dataset. We suggest using the [SmolLM-135M](#) model, which is a small language model with 135 million parameters. The language model should be fine-tuned on the [TinyShakespeare dataset](#).

We will evaluate you on two aspects: fine-tuning and evaluation of the model.

Fine-tuning

You should integrate the language model and the fine-tuning dataset into the provided training framework. Feel free to adapt the training approach as needed to fit whatever compute you have available. You don't need to perform an entire training epoch if you don't have the compute for it. You can train for a few steps and then evaluate the model on a small subset of the dataset. The goal is to demonstrate that you can fine-tune a language model on a small dataset and evaluate its performance.

Evaluation

While it is optional for you to fine-tune a performant LM, we expect you to implement proper evaluation procedures. You can evaluate the off-the-shelf weights from Hugging Face. You should choose and implement appropriate evaluation metric(s).

Hints:

- The training objective is based on next-token prediction. For evaluation, you should identify at least one metric that evaluates the quality of sequence generation.
- Optional: There are automated metrics that evaluate the quality of text generation by comparing to reference texts.

Evaluation Criteria

We will evaluate you on the following aspects:

- Good coding practice: Use the tools included in the template
- Implementation of ML pipeline:
 - We will evaluate the correctness and the quality of the implementation.
 - Considering the limited amount of time you have to complete the task, it's okay if things are not perfect, but you should understand how to improve the code given additional time.
- General understanding of nature language processing (NLP):
 - During the interview, we will ask you to explain your implementation and evaluate your understanding of general understanding of NLP. We do not expect you to have in-depth knowledge of NLP, but we expect you to have a general understanding of how causal language models work and what the training objective and evaluation metrics are.

- Analysis of results:
 - During the interview, we will also ask you to analyze your results and discuss your insights from this exercise.

Deliverables

- After you implement your solution, archive the repo in a zip file and email it to us.
- Make sure to exclude any large files (e.g. the dataset, the SLM model weights, any large, generated results). Please include everything else that's needed to run and evaluate the code, e.g. .git/, .github/, pyproject.toml, etc.